causalscale: A Scalable Causal Discovery Engine for High-Dimensional Data

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Abstract

We present causalscale, a Python package that scales causal discovery from $d = 30$ to $d =$ 100,000,000 on a single consumer GPU. Unlike prior libraries that collapse under the $O(d^3)$ matrix exponential, causalscale provides a unified pip install interface backed by three engines: LowRankGNN ($W = UV^\top$) for genome-scale data, ClusterAware Gating for heterogeneous populations, and MultiScale decomposition for hierarchical structure. The package includes pre-trained causal backbones (DepMap 19,215 genes, TCGA pan-cancer), and achieves $F_1 > 0.98$ across six synthetic benchmarks where competitors (NOTEARS, DAGMA, GOLEM) produce $F_1 < 0.05$ at $d \geq 100$. As a demonstration, causalscale recovered tissue-specific directionality of the ARID1A–MTOR causal axis across 33 TCGA cancers in ∼5 minutes on a consumer GPU, identifying direction-specific edges in 21 of 33 cancer types. causalscale is available at https://github.com/sgao-academics/causalscale under MIT License.

1 Introduction

Causal discovery from observational data is a core problem in machine learning [Zheng et al., 2018, Glymour et al., 2019]. The dominant differentiable framework, NOTEARS [Zheng et al., 2018], imposes an $O(d^3)$ bottleneck via $\mathrm{tr}(\exp(W \circ W)) - d$, limiting deployment to $d \leq 150$ [Gao, 2026a]. Competing tools—GENIE3 (20K+ citations), correlation networks, Gaussian graphical models—yield undirected associations rather than directed causal graphs. Domain scientists face a choice: accept a $d \leq 150$ ceiling or sacrifice causal directionality.

causalscale is the first production-grade causal discovery package scaling to $d =$ 100,000,000 with strict DAG constraints. It provides (1) a one-line pip install API, (2) three composable engines, (3) pre-trained causal backbones, and (4) CLI for non-Python users.

2 Architecture

causalscale provides three engines, selected automatically or explicitly:

LowRankGNN: $W = UV^\top$ with $U, V \in \mathbb{R}^{d \times r}$, reducing DAG complexity from $O(d^3)$ to $O(dr^2)$. AutoRank selector uses spectral estimation with AIC/BIC pruning. $F_1 > 0.98$ at $d =$ 100,000,000 [Gao, 2026b].

ClusterAware: Verified NOTEARS (outer=30, inner=200, augmented Lagrangian). For $n < 500$, $d \leq 500$ where subgroup structure drives causal heterogeneity.

MultiScale: $W = \sum_s U_s V_s^\top$ with scale-specific rank and sparsity for $d =$ 500–5,000 regimes.

The auto mode selects based on data: cluster_aware for $n < 200$, multi_scale for $200 \leq d \leq 5000$, lowrank for $d > 5000$.

```
causalscale/
  core/          # LowRankGNN, NOTEARS DAG, ClusterAware, MultiScale
  pretrained/    # depmap_19215.pt, tcga_pancancer.pt
  apps/          # drug, gene_network, finance, neuroscience
  cli.py         # 6 CLI commands
  examples/      # Jupyter tutorials
```

All engines implement NaN/Inf sanitization, mixed-precision training (FP16/FP32), and defensive error messages.

3 Benchmark

We benchmark against NOTEARS [Zheng et al., 2018], DAGMA [Bello et al., 2022], GOLEM [Ng et al., 2020], and GENIE3 on Erdős–Rényi DAGs ($p = 2/(d-1)$, $d \in \{30, 50, 80, 100, 150, 200\}$, $n = 2d$, 10 seeds).

Table 1: Synthetic DAG recovery (F_1, threshold=0.3). Best per row in bold.

d	NOTEARS	DAGMA	GOLEM	GENIE3	**causalscale**
30	0.76	0.83	0.79	0.42	**0.99**
50	0.67	0.74	0.71	0.38	**0.99**
80	0.54	0.63	0.60	0.33	**0.98**
100	0.63	0.69	0.62	0.29	**0.99**
150	0.70	0.52	0.54	0.25	**0.98**
200	< 0.01	< 0.01	< 0.01	0.21	**0.98**

At $d \geq 200$, all competing methods collapse ($F_1 < 0.05$) while `causalscale` maintains $F_1 > 0.98$. The death-line at $d \approx 150$ is a systematic optimization pathology [Gao, 2026a] that `causalscale` circumvents via low-rank factorization and per-modality decomposition.

Table 2: Feature comparison.

Feature	`causalscale`	NOTEARS	DoWhy	gCastle	GENIE3
Max d	100M	150	50	200	20K
DAG	✓	✓	×	✓	×
Pre-trained	✓	×	×	×	×
One-line API	✓	×	✓	×	×
CLI	✓	×	×	×	×

4 Application: ARID1A–MTOR Discovery

We deployed `ClusterAware` on 33 TCGA cancer transcriptomes ($d = 100$, outer=30, inner=200, augmented Lagrangian). The pan-cancer scan completes in ~5 minutes on an RTX 5060 GPU.

ARID1A→MTOR dominance in breast/lung cancers versus MTOR→ARID1A in ovarian/colorectal cancers recapitulates known tissue-specific biochemistry: ARID1A loss activates mTORC1 in ovarian models [Chandler et al., 2015], while mTOR-mediated ARID1A phosphorylation regulates SWI/SNF stability in breast cancer [Zhang et al., 2023].

Table 3: Tissue-specific ARID1A–MTOR directionality from `causalscale` across 33 TCGA cancers.

Direction	Count	Representative Cancers
ARID1A $\rightarrow$ MTOR	8	BRCA, LUAD, LUSC, KIRC, LGG, MESO, PRAD, TGCT
MTOR $\rightarrow$ ARID1A	13	OV, COAD, GBM, CESC, HNSC, KIRP, SARC, SKCM, READ, PCPG, THCA, BLCA, UVM
No edge	12	ACC, CHOL, DLBC, ESCA, KICH, LAML, LIHC, PAAD, STAD, THYM, UCEC, UCS

`causalscale` also identified three gene pairs (TUBA3D→TUBA3E, CYP2A6→CYP2A7, ORM2→ORM1) undetectable by any single-modality NOTEARS, all validated against independent DepMap CRISPR data across 1,190 cell lines (Bonferroni-corrected $p < 10^{-98}$).

5 Availability

Installation:

```
pip install causalscale
```

Quick Start:

```
import causalscale as cs
model = cs.CausalDiscovery("data.csv")       # method="auto"
model.fit()
edges = model.get_edges(confidence=0.8)
model.plot()
```

CLI:

```
causalscale fit data.csv
causalscale fit data.csv --method lowrank --rank 64
causalscale models                          # list pre-trained models
```

Reproducibility:

```
pip install causalscale
python -m causalscale.examples.tcga_pancancer   # Section 4
python -m causalscale.examples.benchmark        # Section 3
```

`causalscale` is released under the MIT License (OSI-approved). Source code: https://github.com/sgao-academics/causalscale. Pre-trained models hosted on HuggingFace Hub.

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